

Laboratory 3 (100 Points)

Assignment 1 (20 Points)

Design a logic diagram to implement 4-bit ripple carry adder.

1. Design a logic diagram for a full adder.

TABLE 3-12
Truth Table of Full Adder

					Inputs			Outputs	
					X	Y	Z	C	S
Z	0	0	0	0	0	0	0	0	0
X	0	0	1	1	0	0	1	0	1
+Y	+0	+1	+0	+1	0	1	0	0	1
CS	00	01	01	10	0	1	1	1	0
Z	1	1	1	1	1	0	0	0	1
X	0	0	1	1	1	0	0	0	1
+Y	+0	+1	+0	+1	1	0	1	1	0
CS	01	10	10	11	1	1	0	1	0
					1	1	1	1	1

S		YZ			
		00	01	11	10
X	0		1		1
	1	1		1	

C		YZ			
		00	01	11	10
X	0			1	
	1		1	1	1

$$S = \overline{X}\overline{Y}Z + \overline{X}Y\overline{Z} + X\overline{Y}\overline{Z} + XYZ$$

$$= X \oplus Y \oplus Z$$

$$C = XY + XZ + YZ$$

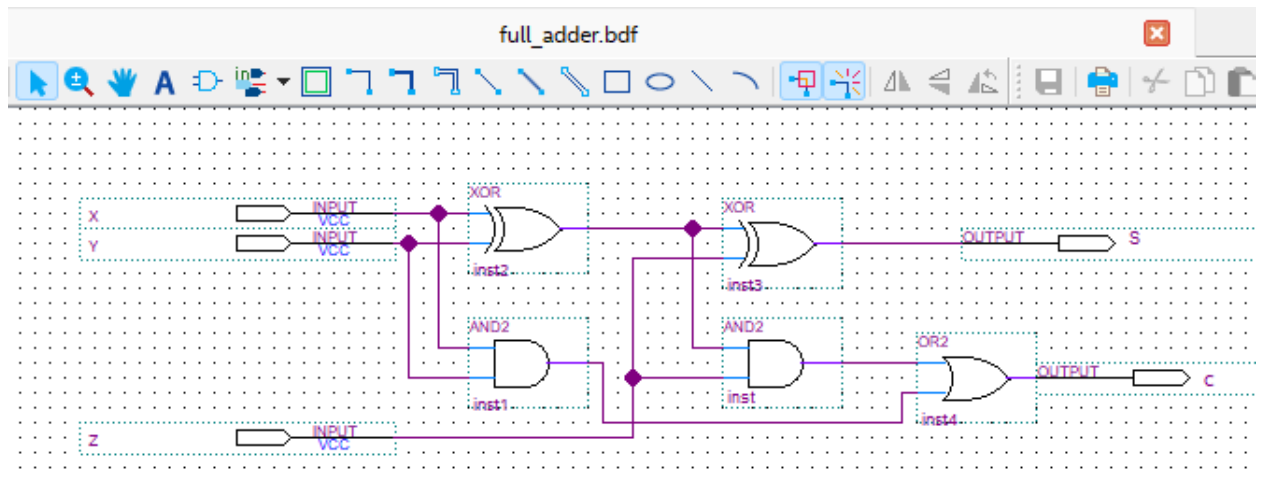
$$= XY + Z(X\overline{Y} + \overline{X}Y)$$

$$= XY + Z(X \oplus Y)$$

35

a. Create a new project named **Adder**.

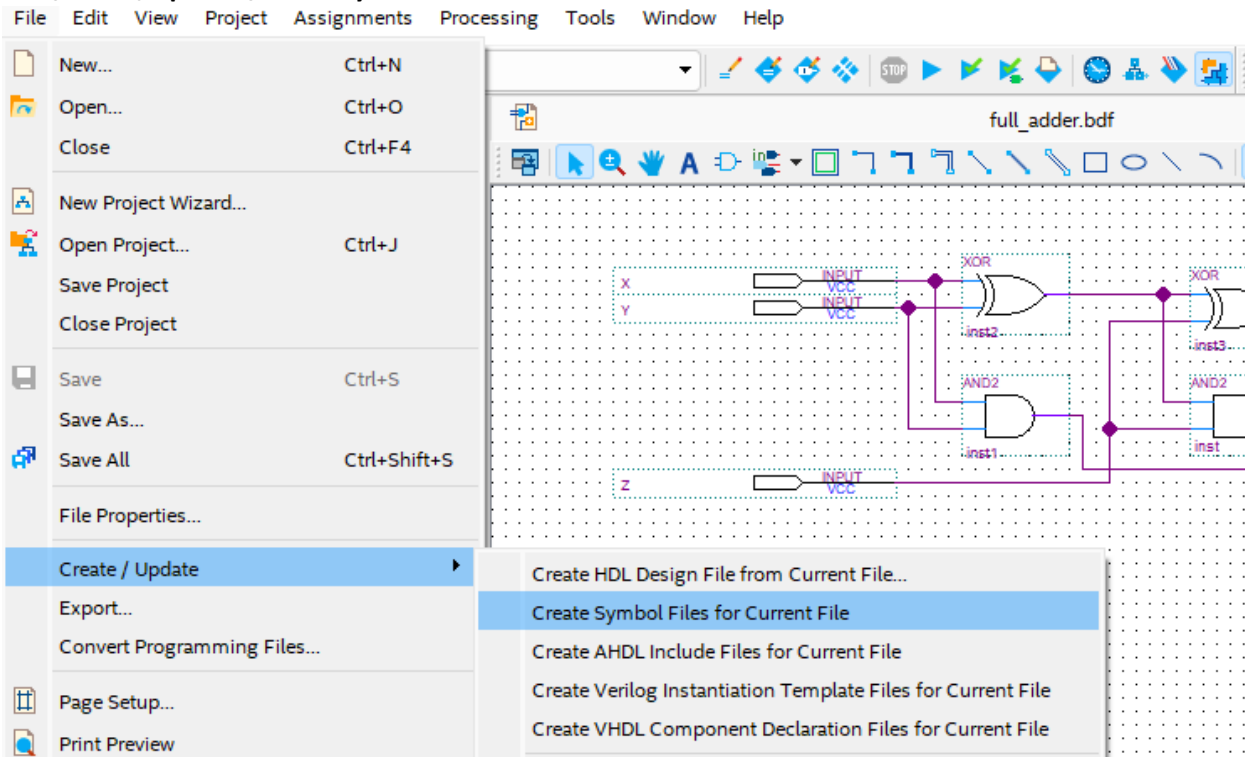
b. Add a new block diagram file (schematic file) named **full_adder.bdf** into the project.

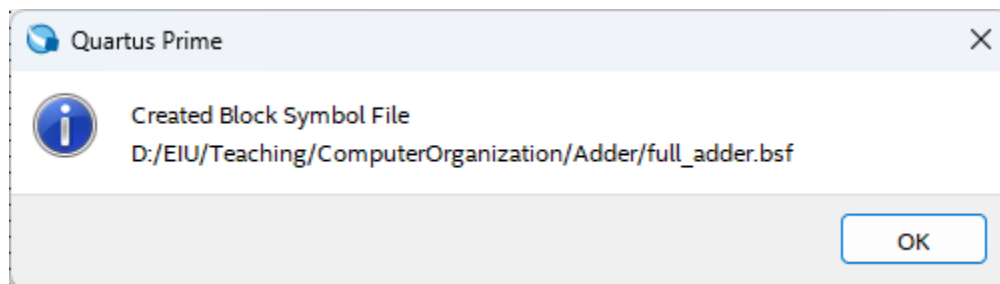
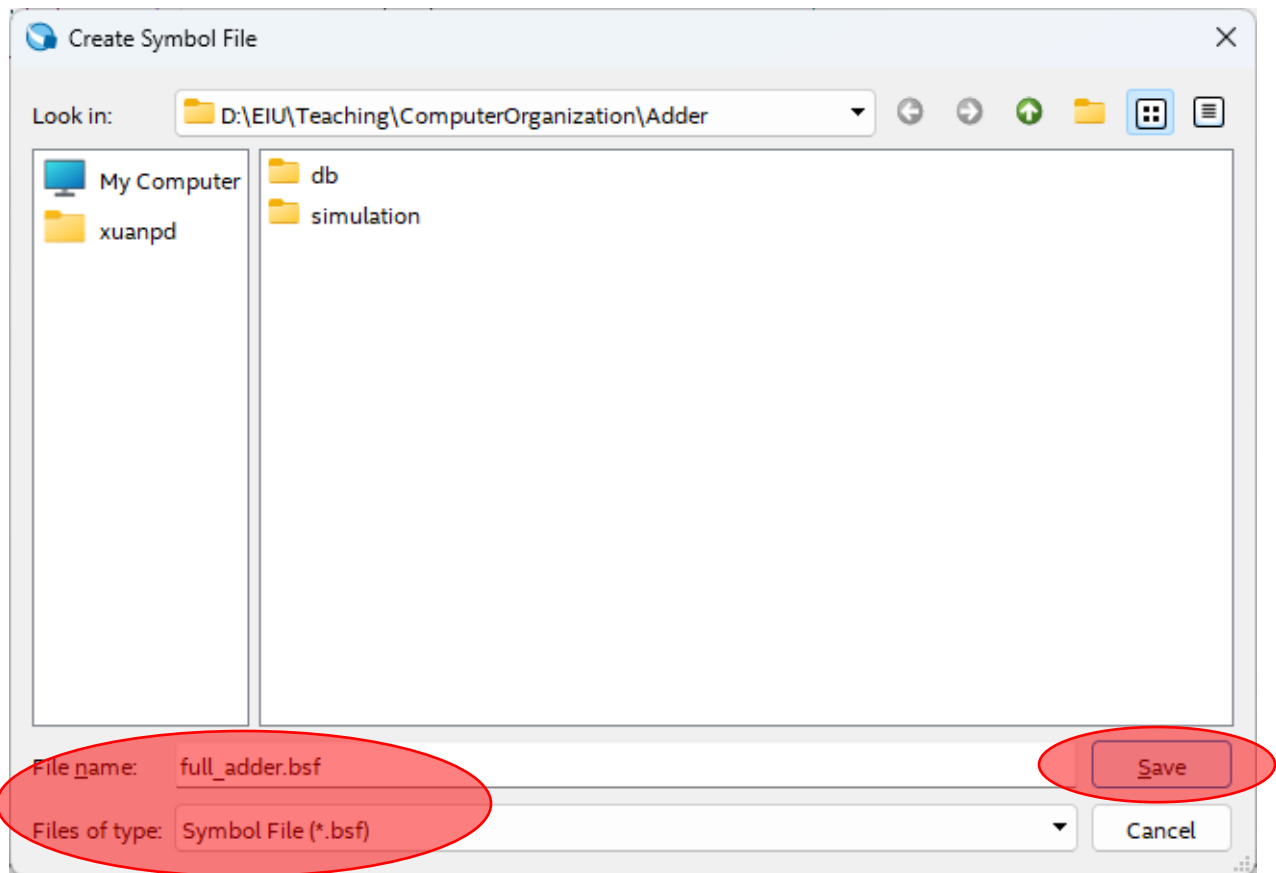


2. Create a symbol for the block diagram **full_adder**.

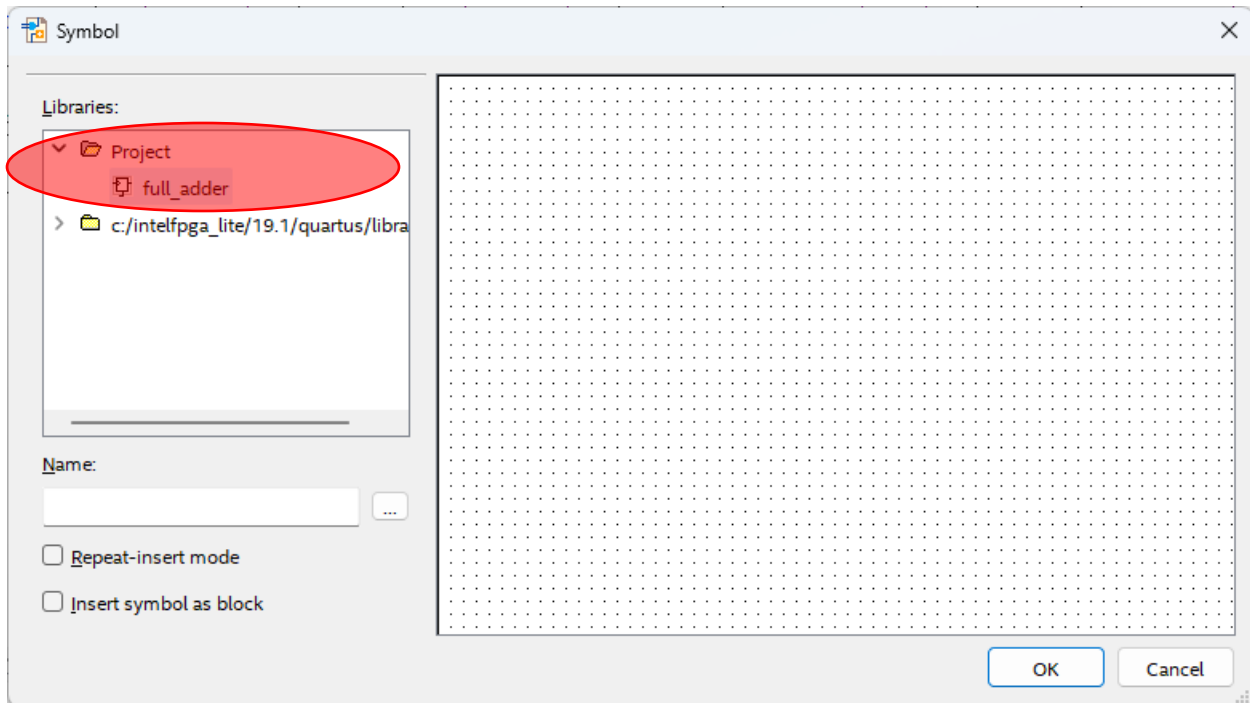
- Make sure that the diagram **full_adder** is currently open. Select

File\Create/Update\Create Symbols File for Current File.

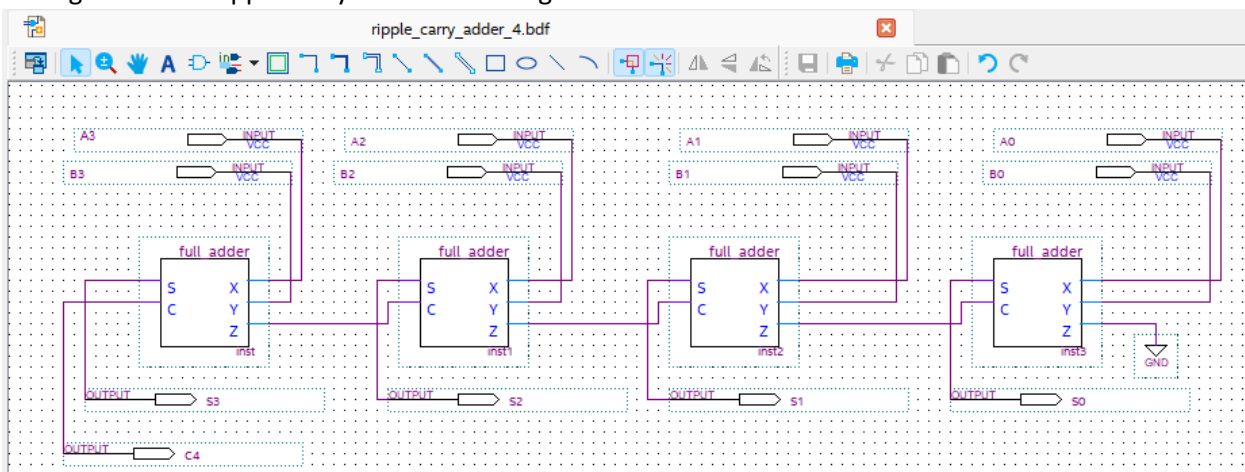




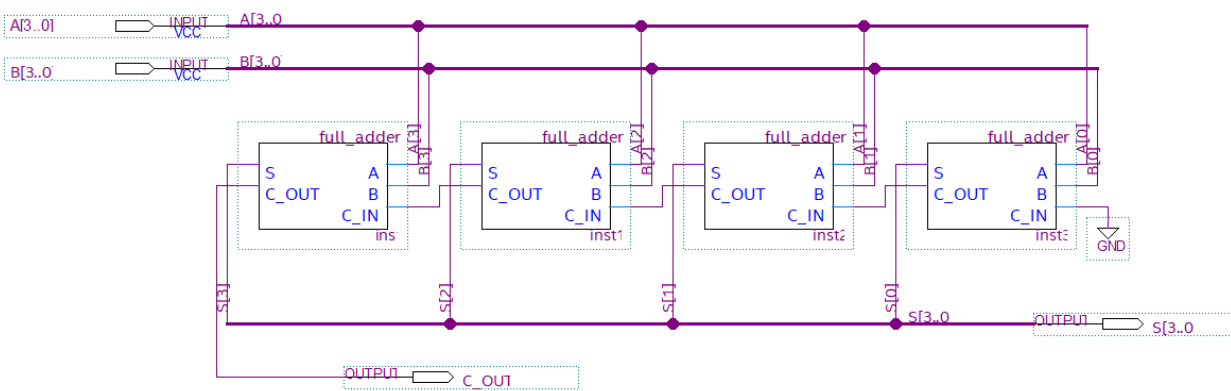
3. Design a block diagram of 4-bit ripple carry adder named **ripple_carry_adder_4.bdf** using **full_adder** circuit created in the previous step.
- Insert **full_adder** symbol:



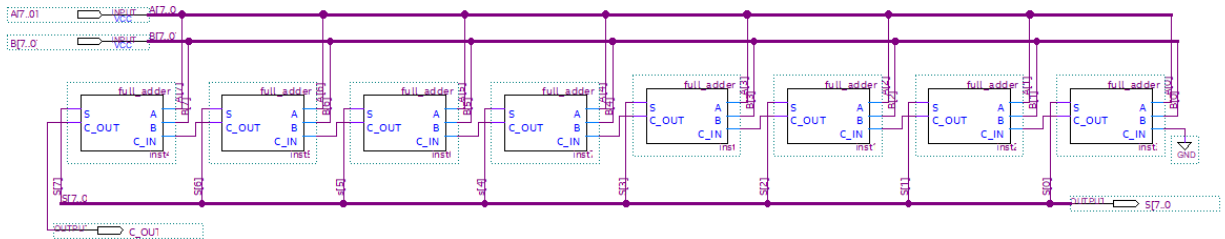
- Design the 4-bit ripple carry adder block diagram



Or: using bus lines



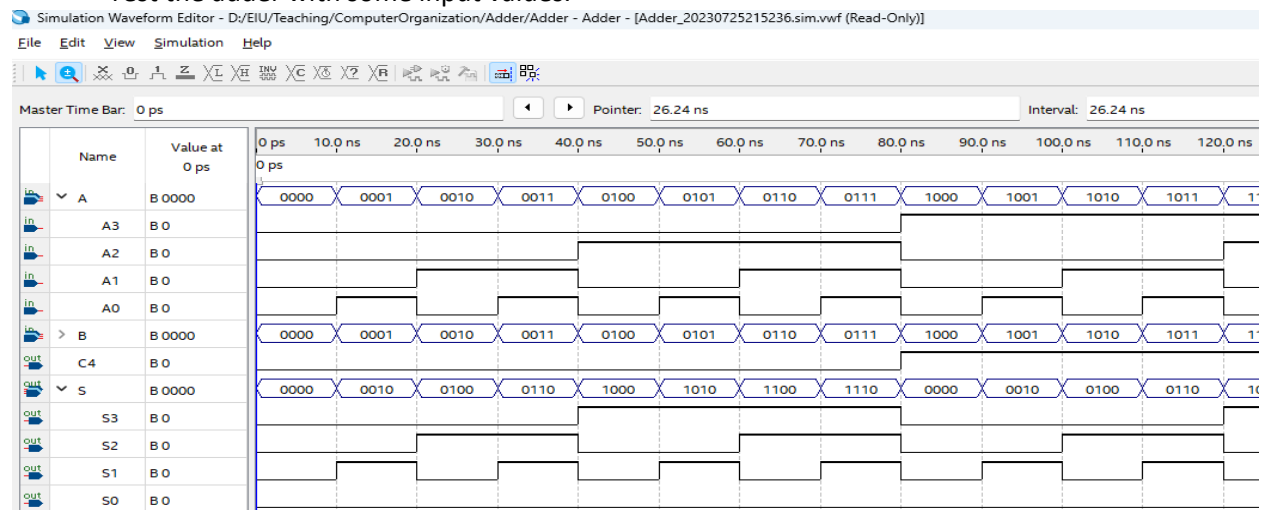
8bits but using bus lines:



- Compile the design

4. Simulate the 4-bit ripple carry adder using waveform editor tool

Test the adder with some input values.



Assignment 2 (20 Points)

Design a logic diagram to implement 4-bit adder-subtractor.

Assignment 3 (Problem 3.3) (20 Points)

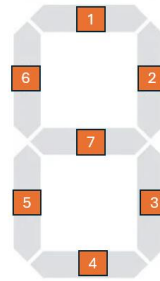
Design a Gray code-to-BCD(Binary-Coded Decimal) code converter that gives output code 0000 for all invalid input combinations. Assume that the Gray code sequence for decimal numbers 0 through 9 is 0000, 0001, 0011, 0010, 0110, 0111, 0101, 0100, 1100, and 1101. All other input combinations should be considered to be invalid.

a. Truth table

Decimal	BCD (Input)				Gray Code (Output)				Decimal Gray Code
	A3	A2	A1	A0	B3	B2	B1	B0	
0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1	1
2	0	0	1	0	0	0	1	1	3
3	0	0	1	1	0	0	1	0	2
4	0	1	0	0	0	1	1	0	6
5	0	1	0	1	0	1	1	1	7
6	0	1	1	0	0	1	0	1	5
7	0	1	1	1	0	1	0	0	4
8	1	0	0	0	1	1	0	0	12
9	1	0	0	1	1	1	0	1	13
10	1	0	1	0	0	0	0	0	0(15)
11	1	0	1	1	0	0	0	0	0(14)
12	1	1	0	0	0	0	0	0	0(10)
13	1	1	0	1	0	0	0	0	0(11)
14	1	1	1	0	0	0	0	0	0(9)
15	1	1	1	1	0	0	0	0	0(8)

Assignment 4 (Problem 3.7) (20 Points)

Given the seven-segment LED shown in the figure, design a logic diagram to generate a seven-segment LED counter from 0 to 9, knowing the truth table as given.



Input				7 segments led						
A3	A2	A1	A0	D7	D6	D5	D4	D3	D2	D1
x	x	x	0	0	0	0	0	0	0	0
0	0	0	0	0	1	1	1	1	1	1
0	0	0	1	0	0	0	0	1	1	0
0	0	1	0	1	0	1	1	0	1	1
0	0	1	1	1	0	0	1	1	1	1
0	1	0	0	1	1	0	0	1	1	0
0	1	0	1	1	1	0	1	1	0	1
0	1	1	0	1	1	1	1	1	0	1
0	1	1	1	0	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	0	0	1	1	1

Assignment 5 (Problem 3.8) (20 Points)

Design a combinational circuit that accepts a 3-bit number and generates a 7-bit binary number output equal to the square of the input

a. Truth table

Inputs			Outputs					
A2	A1	A0	Z5	Z4	Z3	Z2	Z1	Z0
0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0	1
0	1	0	0	0	0	1	0	0
0	1	1	0	0	1	0	0	1
1	0	0	0	1	0	0	0	0
1	0	1	0	1	1	0	0	1
1	1	0	1	0	0	1	0	0
1	1	1	1	1	0	0	0	1